# Jaeho Jeon 전재호 · 田宰昊

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Privacy and Applied Cryptography Lab (PACL) · Advisor: Prof. Young-Sik Kim

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## RESEARCH INTERESTS

Side-channel analysis · Hardware security · Post-quantum cryptography implementation · Cryptographic engineering · Computer architecture · Coding theory

My research focuses on **side-channel analysis**. My goal is to become a uniquely recognized expert in side-channel analysis, and to gradually expand into computer architecture, quantum computing, networking, and coding theory.

## EDUCATION

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| 2026.02 – present | **Ph.D. student**, Electrical Engineering and Computer Science, DGIST |
| 2024.02 – 2026.02 | **M.S.**, Electrical Engineering and Computer Science, DGIST |
| 2017.02 – 2024.02 | **B.S.**, School of Undergraduate Studies, DGIST |
| 2014.03 – 2017.02 | Cheonan Upsung High School, Cheonan |

## PUBLICATIONS

**RB-LDC: Redundancy-Balanced Latent Coding for Robust Diffusion.** Hyunseok Jeong, **Jaeho Jeon**, Young-Sik Kim. *The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026)*, accepted.

**Codeword Masking Can Be Harmful Under Replay Attacks on HQC.** **Jaeho Jeon**, Yongseong Park, Jaeyeon Lee, Suseong Lee, Donghyen Kim, Young-Sik Kim. *International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT 2026)*, accepted. Cryptology ePrint Archive 2026/071.

## HONORS & AWARDS

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| 2026.09 | **Grand Prize, 4th KICS HFR Paper Award** (제4회 KICS HFR 논문대상), Korean Institute of Communications and Information Sciences. Yongseong Park, Hyunseok Jeong, Jaeho Jeon. Total prize $10,000. |
| 2026.06 | **GRAVITY 2026 Startup Competition**, passed the DGIST preliminary round with a hardware-hacking (side-channel) certification service for cryptographic products. Prize $10,000. |

## TEACHING & MENTORING

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| 2026.06 – 2026.09 | **Teaching Assistant**, Pre-URP, DGIST. Mentored a student team on *"Reading the Current to Break Ciphers: Side-Channel Analysis on RSA/AES"*. |
| 2025.03 – 2025.12 | **Teaching Assistant**, UGRP (Undergraduate Group Research Program), DGIST. Supported the team that built an optimized PQC implementation on ARM Cortex-M4. The team won two Grand Prizes: PQC Migration Challenge (NIA, 2025.11) and Undergraduate Best Paper Award, 2025 KICS Fall Conference. |

## ACADEMIC & STUDENT SERVICES

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| 2026.07 | **Staff**, DGIST AI & Crypto Summer Camp 2026 — organized the camp, hosted the rump session |
| 2026.04 – 2026.09 | **President**, DGIST Graduate Student Council — representing graduate students of all departments |
| 2026.03 – present | **Graduate Student Representative**, Dept. of EECS, DGIST |
| 2024.10 | **Treasurer**, 2024 EECS & AI DGIST Student Conference & Happy Hour |

# Student Statement

**Jaeho Jeon** · Ph.D. student, Department of Electrical Engineering and Computer Science, DGIST (Daegu, South Korea) · Privacy and Applied Cryptography Lab (PACL), advised by Prof. Young-Sik Kim · dgwogh@dgist.ac.kr

Presenting: **"Codeword Masking Can Be Harmful Under Replay Attacks on HQC"** (presenting author), accepted to Asiacrypt 2026

## Summary of research interests

My research focuses on **side-channel analysis** and the implementation security of cryptographic systems. I study how secret information leaks from devices through physical channels such as power consumption, and how countermeasures that look sound on paper can fail, or even backfire, when they meet a concrete attacker model. My recent work centers on **post-quantum cryptography (PQC)**, in particular code-based schemes: our Asiacrypt 2026 paper shows that codeword masking, a widely used countermeasure in HQC implementations, can actually be harmful under replay attacks, and analyzes the conditions under which it reduces rather than improves security.

My long-term goal is to become a recognized expert in side-channel analysis and to grow outward from it into computer architecture, networking, and coding theory, so that the security of a cryptographic system can be assessed end to end, from the algorithm down to the silicon.

## How I will benefit from participating in Asiacrypt

Asiacrypt 2026 will be my first time presenting at an IACR flagship conference. I am a first-year Ph.D. student, and I will present this paper on behalf of our team. Presenting it in person to the researchers who design, implement, and attack code-based schemes is the most direct way to test our claims, to learn which assumptions the community finds convincing, and to understand which follow-up questions matter most. Since HQC was selected by NIST for standardization, the question of which countermeasures are safe to deploy has become urgent, and I expect the discussions around our result to shape the next steps of my thesis.

Beyond my own talk, Asiacrypt is where the PQC, implementation-security, and side-channel communities meet. My lab is relatively small and geographically distant from most of these groups, so a few days of conversations at the conference are worth months of reading papers alone. What I learn at Asiacrypt will also be shared with the other members of my laboratory.

I would be grateful for the opportunity to present our work and to take part in the conference in person. Thank you for considering my application.

**Jaeho Jeon** · October 19, 2026

**DGIST · Privacy and Applied Cryptography Lab (PACL)**
Department of Electrical Engineering and Computer Science, Daegu Gyeongbuk Institute of Science and Technology, Daegu, South Korea

October 19, 2026

**To the Asiacrypt 2026 Student Stipend Committee**

**Re: Letter of support and justification of financial need for Jaeho Jeon**

I am writing to support the student stipend application of my Ph.D. student, **Jaeho Jeon**, who will present the paper *"Codeword Masking Can Be Harmful Under Replay Attacks on HQC"* at Asiacrypt 2026 on behalf of our team.

Jaeho joined my laboratory as a master's student in 2024 and began his Ph.D. in February 2026. He works on side-channel analysis and the implementation security of post-quantum cryptography. The Asiacrypt paper, which shows that codeword masking, a common countermeasure for HQC, can be harmful under replay attacks, is a joint work of our laboratory, and Jaeho will present it at the conference as the representative of the team. He is exactly the kind of early-career researcher who benefits most from presenting at, and meeting the community of, an IACR conference.

**Supplementary funding.** The laboratory will cover Jaeho's travel expenses for Asiacrypt 2026 in Hong Kong, including airfare and accommodation. However, we do not have funding available for the conference registration fee, and I would appreciate the committee's support for the registration cost.

I am confident that Jaeho will make full use of the opportunity, and that his participation will benefit both his development and the discussion of countermeasures for code-based PQC at the conference. Please do not hesitate to contact me if any further information is needed.

Sincerely,

**Young-Sik Kim**, Professor
Department of Electrical Engineering and Computer Science, DGIST
Privacy and Applied Cryptography Lab (PACL)
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